Lecture 4: Rules-based chatbots

PSYC 51.07: Models of language and communication

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Assignment 1: Q&A

Questions welcome

Let's take some time to address any questions about Assignment 1:

- Implementation challenges
- Pattern matching strategies
- Configuration file format
- Testing approaches

Remember

There are no bad questions. If you're confused about something, others probably are too.

Common issues and tips

Technical tips

- Use raw strings for regex: `r"pattern"`
- Test patterns on regex101.com
- Handle edge cases (empty input, special chars)
- Print intermediate results for debugging

Common pitfalls

- Forgetting pre-substitutions before matching
- Case sensitivity issues
- Greedy vs. non-greedy matching
- Substitution order matters

Beyond ELIZA

A foundation for more

ELIZA (1966) was just the beginning. Weizenbaum's ideas inspired other researchers to explore what rules-based systems could do.

Today we'll explore:

- **PARRY** (1972): A different kind of simulation
- **A.L.I.C.E.** (1995): Pattern matching at scale
- The limits of rules-based approaches

PARRY (1972)

Further reading

Colby, Weber, & Hilf (1971, *Artificial Intelligence*): Artificial Paranoia

- Created by psychiatrist **Kenneth Colby** at Stanford
- Simulated a patient with **paranoid schizophrenia**
- Different goal than ELIZA: model a specific mental illness
- Had internal state: beliefs, emotional level, goals

Key insight

ELIZA *reflects*; PARRY *models*. ELIZA avoids commitment; PARRY has a coherent (if paranoid) worldview.

PARRY vs ELIZA

ELIZA (1966)

- Non-directive therapy
- Reflects user input back
- No internal state or beliefs
- Avoids making claims
- Goal: keep user talking

PARRY (1972)

- Simulates a patient
- Has beliefs about the world
- Tracks emotional "anger" level
- Makes paranoid claims
- Goal: behave like a specific illness

Important caveat

Both are still rule-based systems with no real understanding. The "beliefs" in PARRY are just variables that affect pattern selection.

How PARRY works: The algorithm

PARRY uses a **state machine** with emotional variables:

```python
class Parry:
    def __init__(self):
        self.anger = 5          # 0-20 scale
        self.fear = 8           # 0-20 scale
        self.mistrust = 10      # 0-15 scale
```

Processing steps

1. **Match pattern** against input (like ELIZA)
2. **Update emotional state** based on topic
3. **Select response** influenced by anger/fear/mistrust levels
4. **Apply threshold rules** for extreme reactions

PARRY example: Step by step

Input: "Tell me about the mafia"

Step 1: Pattern Match

```
/\b(mafia|mob)\b/i → MATCH
```

Step 2: Emotional Update

```
fear += 4    → 8 → 12
anger += 3   → 5 → 8
mistrust += 3 → 10 → 13
```

Step 3: Response Selection

High fear + high mistrust triggers paranoid responses:

```
"You know, they have their
 ways of getting to you."
```

Step 4: State Carried Forward

Next input processed with elevated emotional levels.

PARRY's emotional dynamics

```
Low emotions → Cooperative responses
    "I used to gamble on horses."
    "I've been feeling tense."

Medium emotions → Guarded responses
    "I don't want to talk about that."
    "Why are you asking me this?"

High emotions → Paranoid/hostile responses
    "Are you one of THEM?"
    "You're trying to trick me!"
```

Key insight

The emotional state creates **coherent behavior over time**. ELIZA has no memory; PARRY's responses depend on conversation history.

The Turing Test, revisited

A famous experiment

In 1972, PARRY was tested via teletype against real patients and psychiatrists. Judges could not reliably distinguish PARRY from actual patients with paranoid schizophrenia.

- This was one of the first informal "Turing tests"
- Success? Or a comment on how we judge understanding?
- Psychiatrists were looking for *symptoms*, not *understanding*

Think about it

Does passing a specialized test mean the system understands anything? What does it mean that experts could be fooled?

A.L.I.C.E. (1995)

The AIML era

A.L.I.C.E. (Artificial Linguistic Internet Computer Entity) was created by Richard Wallace.

Key innovations:

- **AIML** (Artificial Intelligence Markup Language)
- Over **40,000 patterns** (vs ELIZA's ~200)
- Won the Loebner Prize three times (2000, 2001, 2004)
- Open source, widely studied and extended

Note

More patterns $\neq$ more understanding. A.L.I.C.E. is still fundamentally pattern matching.

AIML: Pattern matching at scale

```
<category>
  <pattern>MY NAME IS *</pattern>
  <template>
    Nice to meet you, <star/>.
  </template>
</category>

<category>
  <pattern>I AM FEELING *</pattern>
  <template>
    Why are you feeling <star/>?
  </template>
```

Sound familiar?

This is the same decomposition/reassembly pattern as ELIZA, just in XML format with more extensive coverage.

How ALICE works: The algorithm

AIML Processing Steps:

1. **Normalize input**: "Don't you think so?" → "DO NOT YOU THINK SO"
2. **Find matching pattern**: Search through 40,000+ patterns
3. **Extract wildcards**: `*` captures arbitrary text
4. **Process template**: May include conditionals, `<srai>` redirects
5. **Generate response**: Substitute captured text

Key difference from ELIZA

AIML supports **recursive processing** via `<srai>` (Symbolic Reduction AI), allowing patterns to trigger other patterns.

ALICE example: Step by step

Input: "My name is John and I like pizza"

Step 1: Normalization

```
MY NAME IS JOHN AND I LIKE PIZZA
```

Step 2: Pattern Search

```
<pattern>MY
```

Step 3: Template Processing

```
<template>
  Nice to meet you, <star/>.
  <think> <set name="name">
```

AIML advanced features

```xml
<!-- Symbolic Reduction (SRAI) -->
<category>
  <pattern>HI THERE</pattern>
  <template><srai>HELLO</srai></template>
</category>

<!-- Topic-based context -->
<topic name="MOVIES">
  <category>
    <pattern>WHAT DO YOU LIKE</pattern>
    <template>I enjoy science fiction films.</template>
  </category>
```

Still rule-based

Despite these features, ALICE cannot generalize beyond its patterns. 40,000 rules still miss infinite valid inputs.

Live demo

Interactive exploration

Let's interact with these historical systems:

Chatbot Evolution Demo

While exploring, consider:

- How do PARRY and A.L.I.C.E. differ from ELIZA?
- What do they do surprisingly well?
- Where do they break down?
- Can you find the edges of their rule sets?

Discussion: The limits of rules

Discussion questions

1. More rules $\neq$ understanding. Why not?
2. What *can't* rules-based systems do?
3. Where do they break down?
4. Is there a fundamental limit, or just a practical one?

Consider these failure modes:

- Novel situations not covered by rules
- Context that spans multiple turns
- Reasoning about cause and effect
- Understanding sarcasm, irony, metaphor

The evolution of chatbots

ELIZA (1966) → PARRY (1972) → A.L.I.C.E. (1995) → Modern LLMs

From hand-crafted rules to learned patterns

Note

The trajectory: fewer hand-crafted rules, more learned patterns. But the fundamental question remains: does *any* of this constitute understanding?

What rules-based systems teach us

1. **Behavior ≠ understanding**
    - A system can *seem* intelligent without understanding anything
    - This is the lesson of the Chinese Room
2. **Scale is not a solution**
    - 40,000 patterns is not qualitatively different from 200
    - More rules just mean more edge cases
3. **Pattern matching can fool us**
    - We project understanding onto systems that lack it
    - The ELIZA effect applies to all of these systems

Weizenbaum's legacy

Further reading

Weizenbaum, J. (1976). *Computer Power and Human Reason: From Judgment to Calculation.* W.H. Freeman.

After creating ELIZA, Weizenbaum became a critic of AI:

- Warned against replacing human judgment with computation
- Argued some tasks *should not* be automated
- Foresaw many issues we face today with AI

> *"What I had not realized is that extremely short exposures to a relatively simple computer program could induce powerful delusional thinking in quite normal people."*

Weizenbaum's warnings

His concerns (1976)

- Mistaking simulation for understanding
- Replacing human relationships
- Trusting AI with decisions it cannot make
- Questions about human identity

Still relevant today

- Chatbots as therapists
- AI companions and relationships
- Automated decision systems
- Debates about AI consciousness

50 years later

Weizenbaum's warnings feel remarkably prescient. Are we heeding them?

Next week: Computational linguistics

Week 2 preview

We're leaving hand-crafted rules behind. Next week:

- **Tokenization**: Breaking text into meaningful units
- **Preprocessing**: Cleaning and normalizing text
- **Learning patterns from DATA** instead of writing them by hand

Hand-crafted rules → **Learning from data**

The paradigm shift that enables modern NLP

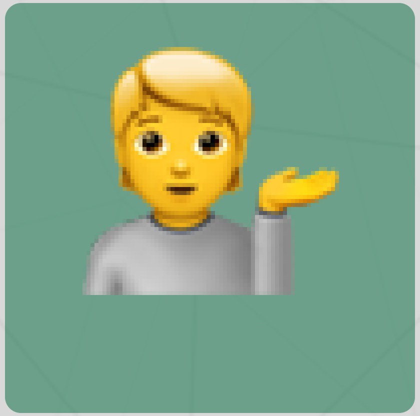
Key takeaways

1. **ELIZA inspired others:** PARRY and A.L.I.C.E. built on Weizenbaum's ideas
2. **Different goals, same method:** Whether reflecting or modeling, all used pattern matching
3. **Scale doesn't solve understanding:** 40,000 patterns is not intelligence
4. **Behavior fools us:** We attribute understanding where none exists
5. **The warnings remain relevant:** Weizenbaum's concerns apply even more today
6. **The future is learning:** Moving from rules to data-driven approaches

Questions? Want to chat more?

Email me

Join our Discord



Come to office hours

> **Tip**
>
> Start working on Assignment 1 now if you haven't already. Reach out early if you get stuck.